

On the Timing and Execution of Open Market Repurchases*

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On the Timing and Execution of Open Market Repurchases

Little is known about the timing and execution of open market repurchases. U.S. firms are under no obligation to disclose when they are trading, and generally report only quarterly changes in shares outstanding. We use 64 firms' supplementally disclosed repurchase trading data to provide the first examination of repurchase timing and execution. Across the days reported in our sample, firms adopted a variety of execution styles ranging from immediate intense repurchasing to delayed and smoothed repurchasing. We find no clear evidence that repurchases are timed to coincide with, precede, or follow, days on which information is released. Within a given repurchase day, we find u-shaped repurchase patterns for NYSE firms and end-of-day repurchasing tendencies for Nasdaq firms. We benchmark the costs and value of a given repurchase program against naive accumulation strategies achieving the same terminal portfolio. While there is considerable variation across the firms, NYSE firms on average beat their benchmarks whereas Nasdaq firms do not. Finally, we document the liquidity impact of open market repurchases. We find that repurchasing contributes to market liquidity by narrowing bid-ask spreads on days when repurchase trades are completed.

1 Introduction

Announcements of U.S. open market repurchase programs have grown dramatically from 131 companies repurchasing \$15.8 billion in 1985 to 1,445 companies announcing repurchases valued at \$149.6 billion in 1999. The importance of stock repurchases as a method of returning capital to shareholders is well established. Previous research has extensively analyzed the volume of repurchase activity, the information content of repurchase announcements and the long-term performance of repurchasing firms. (For example see Jagannathan, Stephens, and Weisbach (2000), Ikenberry and Vermaelen (1996), and Ikenberry, Lakonishok, and Vermaelen (1995)). However, little is known about how U.S. firms execute actual trades during an open market repurchase program.¹ U.S. repurchase execution data are difficult to obtain. Current disclosure standards, as regulated by the FASB and the SEC, impose no obligation on firms to disclose trade execution details.² Quarterly 10Q and annual 10K filings provide the only indication of repurchase activity. Consequently, little is known about the daily timing of repurchase programs or the impact they have on day-to-day trading. This paper provides the first in-depth examination of daily and intra-daily repurchase execution behavior. Using voluntarily disclosed data, we examine the timing and intensity of 64 firms' repurchase programs during 1993 and 1994.

Many theories have been put forth to explain management's use of repurchases as a means of distributing capital to shareholders. For example, Elton and Gruber (1968) suggest that using repurchases (as opposed to dividends) allows each stockholder to set his own current income, making repurchases a preferred means of distributing cash. It has also been argued that because of IRS rules, repurchases have favorable tax advantages relative to dividends. Miller and Scholes (1978) conclude, however, that "the effect of taxes on the optimal financial policies of firms are far less obvious than the simple comparison of tax differentials might seem to suggest." Bagwell (1991) shows that repurchases can be an effective takeover deterrent. It has also been suggested that repurchases are simply substitutes for dividends. Jagannathan,

¹Ikenberry, Lakonishok, and Vermaelen (2000) examine monthly repurchase timing for a set of Canadian firms and find that repurchase activity is linked to price movements. Brockman and Chung (2000) provide some evidence that on the Hong Kong stock exchange, where almost immediate public disclosure of repurchases is required by law, repurchasing firms appear to exhibit timing ability.

²Although the SEC provides safe harbor Rule 10b-18 to protect repurchasing firms from charges of market manipulation, the regulation does not mandate specific disclosure. See Cook, Krigman, and Leach (2000) for an examination of repurchasing firms' use of Rule 10b-18.

Stephens, and Weisbach (2000) demonstrate, however, that repurchases contain the added benefit of financial flexibility relative to dividends and argue that firms do not use them as substitutes.

While each of the many motivations has merit, information signaling has received the most support in the empirical literature.³ Wansley, Lane, and Sarkar (1989) surveyed CFOs of large U.S. corporations about their reasons for repurchasing shares and find, consistent with a signaling motivation, that the most frequently cited reasons for repurchasing are (1) management's belief that the stock is undervalued, and (2) as a signal to investors of confidence in the future level of earnings and stock prices. In line with their findings, the most frequently cited reason for repurchasing stock by the firms in our sample is that they feel that the stock is undervalued.⁴

In a signaling framework, management views the stock as undervalued and announces a buyback program. The announcement is taken as a signal of management's assessment of undervaluation. If the market responds efficiently, prices should instantaneously adjust in an unbiased manner. However, Ikenberry, Lakonishok, and Vermaelen (1995) demonstrate that prices respond slowly to the signal, consistent with market skepticism of the repurchase announcement. A slow adjustment allows firms to repurchase stock at undervalued prices. We examine whether the firms in our sample beat several naive accumulation strategies and thus, earn trading profits on their repurchase programs.

If repurchases are motivated by information signaling, theory suggests that allowing firms to repurchase stock in the open market may be a cause for concern. The microstructure literature identifies two implications of the presence of an informed trader in the market. First, Glosten and Milgrom (1985), among others, suggest that an increased level of informed trading widens the bid-ask spread and increases trading costs. Barclay and Smith (1988) examine annual spread data and conclude that such a widening occurs during repurchase programs. They argue that open market repurchases carry a hidden cost in the form of decreased liquidity. Secondly, Kyle (1985) demonstrates in a dynamic setting that informed

³For example see Vermaelen (1981), Dann (1981), and Asquith and Mullins (1986) among others.

⁴Chan, Ikenberry, and Lee (2000) examine whether managers knowingly repurchase undervalued shares by examining insider purchases and sales surrounding the announcement of open market repurchase programs. They find mixed results. However, a comparison of firms that actually repurchase shares to those that announce a program and repurchase no shares, reveals superior long-term performance for the actual repurchasers, consistent with managers' recognizing undervaluation and taking advantage of it. We take this analysis one step further. We examine the within-program timing ability of managers.

traders may strategically time their trades to maximize profits at the expense of uninformed traders. It remains an open question whether or not corporations are informed traders. That is, do they use their superior information advantageously? By examining execution skill during repurchases, we can gain some insight into whether corporations act as informed traders.

Given that a firm has decided to initiate a repurchase program, the choice of how to execute the program can be motivated by different short term goals and have different empirical implications. For example, a firm may wish to provide liquidity, support a falling stock price, or simply buy shares as cheaply as possible. There is a sizeable literature on the liquidity effects of open market repurchases. For example, Barclay and Smith (1988), Singh, Zaman, and Krishnamurti (1994), and Miller and McConnell (1995) all examine the change in the bid-ask spread surrounding repurchase announcements and find mixed results. We examine spread changes on actual repurchase days to detect liquidity impacts.

If firms are repurchasing with a liquidity or price support motive, the structure of equity trading can affect whether a program can meet its short-term objective. Specifically, how and when firms repurchase may depend upon the exchange where a firm's stock is listed; a specialist system such as the NYSE may afford different opportunities for the firm than the dealer structure used on the Nasdaq. The ability to post limit orders (available on the NYSE) may lower the costs of a repurchase program.⁵ As a few firms in our sample indicated, trading structure can be a relevant consideration.

When we consider how much of a day's volume a firm repurchases, trading protocol may also influence repurchase costs. For example, due to dealer-to-dealer trading on the Nasdaq, reported trading volume is overstated relative to NYSE volume. Accordingly, percentage daily volume acquired is understated (relative to NYSE). To allow for potential microstructure effects, we segregate our sample into two groups – 24 NYSE firms and 41 Nasdaq firms.

The main results we present are:

- Repurchase program executions vary widely across firms. That is, program length, aggression and timing vary widely in our sample. For example, one firm completed its announced program on one trade day, while another repurchased on 288 of 387 trading

⁵At the time of the repurchases in this study, it was not possible to post limit orders effectively on the Nasdaq.

days following the program’s announcement.

- We find evidence that firms decrease repurchasing activity around firm-specific information announcements. Thus they appear to take care to avoid trading on short-term information.
- We find that firms repurchase following price drops, consistent with a price support motivation. On the NYSE, repurchase intensity is related to contemporaneous and lagged price changes and trading volume. Nasdaq firms’ repurchase intensity is related to lagged excess return measures but appears unrelated to contemporaneous price changes.
- Trade execution appears to be influenced by market structure (specifically, the use of limit orders) and firm size. On the NYSE, 40.7% of repurchase trades are executed at the prevailing market bid, consistent with the use of a limit order. On the Nasdaq, only 7.6% of trades are similarly executed.
- We show that repurchase trading contributes to market liquidity through narrower bid-ask spreads on repurchase days than on non-repurchase days.
- From a cost perspective, we find that NYSE firms pay significantly less compared to three different *naive* accumulation strategies. In contrast, Nasdaq firms pay more compared to these same strategies.

The remainder of the paper is as follows. Section 2 describes the solicitation and construction of our repurchase dataset. Section 3 provides a detailed description of several programs in our sample. Section 4 documents daily and intradaily repurchase patterns. Section 5 discusses the methods and benchmarks we use to measure timing ability within the repurchase period. Section 6 examines the liquidity aspects of repurchase programs and Section 7 concludes.

2 The Repurchase Data Set

Data on the details of repurchase execution are not readily available. We directly solicited repurchase data from the 478 firms identified in Securities Data Company’s (SDC) Mergers

and Acquisitions Database as announcing open market repurchase programs between March 10, 1993 and March 4, 1994. Appendix A provides a copy of the questionnaire we sent to the 478 firms. Sixty-eight firms responded to our request. Four of the 68 indicated that they were unable, or unwilling, to provide the requested data. The remaining 64 firms constitute the sample used in our study.⁶ One firm changed its listing from the Nasdaq to the NYSE during its repurchase program and, consequently, our final sample includes 24 NYSE programs and 41 Nasdaq programs.

Firms provided the target dollar amount of the program or the target number of shares, the program status (complete or on-going), the agent designated to coordinate the repurchase activity, and the name of a contact person. The disclosed trading data include the trading dates, the exchange used, the number of shares acquired, the price paid per share, and an indication of whether the prices were exact or a daily average. We undertook several checks to assure the accuracy of the response data. Initially, we checked the reported volume against total daily volumes from the *NYSE Transactions and Quotations (TAQ)* dataset. We contacted the corporations for clarification when we encountered infeasible trades.⁷ Our final dataset includes 84.2 million shares traded on 2,190 firm-trade days for 64 firms in 1993 and 1994.

Table 1 provides summary statistics on the firms in our sample. Our sample represents a wide cross-section of the market. The 1993 average daily trading volume ranges from 32,500 shares to over 95 million shares, with an average of 12.6 million shares. Their market capitalization ranges from \$6 million to over \$31 billion. The percent of the total daily volume acquired in one day ranges from .03% to 100%.⁸ The percent of outstanding shares targeted ranges from .56% to 32.8%, with an average of 7%. The trading frequency varies widely across firms. One firm repurchased their announced program target on one day (a very small program), whereas another firm repurchased on 288 of 387 consecutive trading

⁶Many of the participating corporations provided their repurchase data with the understanding that their participation in the study be confidential.

⁷The most common inconsistencies were due to reporting settlement rather than execution dates, and including brokerage commissions in the price. We exclude eight transactions from six firms which were not feasible and no resolution was found, forty-two reported repurchase days in 1992 (due to the absence of TAQ data), and eight trades reported as private transactions.

⁸SEC regulation 10b-18 restricts non-block repurchase volume to 25% of the average daily trading volume during the 4 calendar weeks prior to the week of a repurchase. Thus, there is no prohibition on taking 100% of a day's total trading volume. For complete details of repurchase regulation and Rule 10b18, see Cook, Krigman, and Leach (2000).

days. On average, 34.5% of the trading days in a firm's reported period contain a repurchase execution. The length of the reported repurchase horizons range from one to 387 days.

2.1 Potential Survey Nonresponse Bias

Gathering data through survey introduces a potential self-selection bias in firms that choose to respond. There are at least two reasons why a firm might not wish to have its repurchase record revealed. First, if a firm is successful in using information to its advantage while repurchasing, accusations of manipulation or fraud may arise. For this reason, it may be difficult to detect repurchase-related excess trading profits in the records of our responding firms. On the other hand, if a firm loses money on its repurchase program, paying too much for its shares or consistently buying at the worst possible times, it may likewise choose not to respond to our survey. Consequently, our dataset most likely contains firms that do not perform too well or too poorly, and our findings are likely attenuated relative to non-responding firms.

We investigate the potential nonresponse bias using two types of data. First we examine firm-level characteristics of responding and non-responding firms. Second, to address potential performance-related reasons for a firm's not wishing to have its repurchase record revealed, we examine market-based return data. Table 2 provides details. The responding firms have a significantly larger mean market value of equity compared to non-respondents (\$2,646 versus \$781 million). However, the data are highly skewed. Non-parametric tests reveal no difference in median firm size. Program size, expressed as a percentage of outstanding shares is similar (6.9%) for responding and non-responding firms. The response count for Nasdaq firms (41 firms or 17.5% of those solicited) is higher than that for NYSE firms (24 firms or 12.1% of those solicited). The higher percentage of responding Nasdaq firms helps to explain the lower annual trading volume we report for our sample of responding firms.

We examine cumulative abnormal returns surrounding the program announcements and for three years following the program announcement for the full set of firms announcing open market repurchase programs during 1993 and 1994. Cumulative abnormal returns at the announcement are measured for two days (0, +1) using a market model. Market model parameters are estimated over the period -100 to -11. Three year abnormal returns are measured two ways. First, we calculate market-adjusted holding period returns relative to the

CRSP value-weighted index. Second, following Lyon, Barber, and Tsai (1999), we calculate holding period size-adjusted returns relative to NYSE size portfolio deciles. Consistent with Ikenberry, Lakonishok, and Vermaelen (1995), we find positive abnormal returns at the announcement and for the three years post announcement. Importantly, we find no difference in returns between the responding and the non-responding firms. Overall, while our sample is small, on these limited dimensions it appears representative of open market repurchase programs in general during the period we study.

The use of survey data can always be subject to criticism, even when the survey data are verifiable (as with repurchase trades). However, we feel that it is important to weigh the potential gain in our understanding of issues not currently addressable with publicly available datasets against any possible over-generalizations fostered by potential self-selection bias. Without an SEC- or FASB-directed change in repurchase disclosure requirements, survey is the only technique available for examining the timing and implementation of U.S. repurchase programs. Ultimately, the possibility of self-selection bias remains and all results should be interpreted with this in mind.

3 Characterizing Repurchase Trading

The level of detail in our repurchase sample permits an examination of questions not previously addressed. Do firms repurchase an equal number of shares on a daily basis throughout a program? Do they repurchase in proportion to market volume? Is the choice of repurchase day related to price movement or general market activity? Do firms time their trades with respect to either information releases or inside information? The firms in our sample exhibit great diversity in trading execution. Therefore, prior to treating our sample as observations from a homogeneous population, we provide a detailed characterization of four different programs that highlight the heterogeneity in our sample. Figure 1 displays the daily repurchase volume of the four firms across time.

- **Firm 1** announced an intent to repurchase 2,100,000 shares in the open market. It waited two days and then repurchased 225,000 shares over the following 6 days, after which it did not successfully repurchase for the next 59 trading days.⁹ The firm then repurchased 1,000,000 shares over 13 of the next 16 trading days. It did not repurchase for the following 25 trading days and achieved its stated target by buying 875,000

⁹As we have data only on successful repurchase executions, we cannot say whether firms were active in the market, say with limit orders, even though not in a way that resulted in actual repurchase.

shares over the subsequent 9 trading days. Over the duration of the program, the firm repurchased between 5,000 and 191,700 shares per day.

- **Firm 2** announced an intent to repurchase up to \$500 million of its common stock in the open market. It waited 10 trade days to enter the market and then executed repurchases on 45 of the following 47 trade days, averaging 210,980 shares per day, and ranging from 50,000 to 1,258,000 shares per day. Two days after this buying period, the firm released an earnings announcement stating “better than expected first quarter results.” On the fifth day following this announcement, the firm resumed repurchasing. It completed its stated target in the next 11 trading days, with average repurchase of 208,590 shares per day.
- **Firm 3** announced an intent to purchase common stock with aggregate market value of up to \$50 million. The firm waited five weeks to enter the market. After repurchasing 513,800 shares over 25 weeks, the firm announced a continuation of the repurchase program (for an additional \$50 million). Over the entire 65 week period of our analysis, the firm purchased a total of 3.6 million shares valued at \$100,000,000 including commissions on 288 of 325 trade days. Daily shares repurchased ranged from 100 to 50,000 with a mode of 6,000 shares per day. The firm’s repurchase behavior does not appear to change for information releases. It also appears unrelated to changes in non-repurchase trading volume.
- **Firm 4** announced an intent to repurchase up to \$25 million of common stock. It waited two days to begin repurchasing. The firm bought 300,000 shares on nine of the following 18 trade days. After 24 weeks without a repurchase, it bought 930,000 shares over 18 of 45 trading days. There appears to be some relationship between the firm’s repurchasing behavior and stock prices during the repurchase period. For example, prior to the announced repurchase, the stock fell from a high of \$15.625 to \$8.75. The firm bought shares in the range of \$8.75 to \$9 per share and stopped buying as the stock price began rising. During the 24 week window when the firm did not trade, the stock traded in the \$12 range. When the price fell to \$9 per share the firm re-entered the market for more repurchases. Following the last reported repurchase, the stock was trading in the \$13 range.

The diversity of these examples suggests that general statements about repurchase program execution can be misleading. By design, any statistical analysis will deemphasize the diversity of styles and objectives of repurchasing firms.

4 Frequency & Timing of Repurchase Trading

The first question we address is “When do firms repurchase and do they represent a significant presence in the market?” Table 3 provides descriptive statistics on the daily timing of repurchases.¹⁰ The average program length for 24 NYSE programs is 195 days, with 66

¹⁰Due to differences in the trading protocol and permissible order types on the NYSE (a specialist system with a limit order book) and the Nasdaq (a dealer system with multiple quotes and no limit order book at the time of our study), we partition our analysis by exchange listing.

days of repurchasing. Repurchases account for an average of 19.5% of trading volume on the days firms execute trades. In aggregate, the NYSE firms repurchased 70.2 million shares, representing 6.6% of total trading volume during the reported repurchase period. On the Nasdaq, repurchase trading accounts for 33% of a repurchase day's reported trading volume. Nasdaq firms, however, only repurchased on an average of 16 out of 116 reported days. The aggregate reported repurchase of 14.6 million shares represents 10.4% of total trading volume during the reported repurchase horizon. That is, about 1 in every 10 shares traded during the repurchase period is a repurchase. Overall, trading on a NYSE repurchase day is more likely (than trading on a Nasdaq repurchase day), although within such a day meeting an NYSE firm as a counterparty is less likely (than meeting a Nasdaq firm as a counterparty).

We also examine trading by the day of the week. Gibbons and Hess (1981), among others, document that Mondays have lower returns than other days of the week. Lakonishok and Maberly (1990) demonstrate that while institutions are less likely to trade on Mondays, individuals are more likely to sell than buy on Mondays. Thus, they find an imbalance between buy and sell transactions which results in slightly lower stock prices on Mondays. A cost minimization objective would suggest increased repurchase activity on Mondays relative to other days of the week. However, consistent with lower institutional trading on Mondays, the NYSE firms in our sample trade slightly less frequently on Mondays than on other days (295 out of 1577 trade days, or 18.7% of the time repurchases occur on Monday). On the Nasdaq, we find that firms appear to trade slightly more on Thursdays and Fridays than on other days of the week. Overall, we do not find evidence that repurchasing firms increase buying activity on Mondays to take advantage of order imbalances and lower stock prices.

Within a trading day, previous research has documented u-shaped patterns in bid-ask spreads and trading volume for Nasdaq and NYSE firms.¹¹ We next examine whether there is any evidence that the firms in our sample exhibit buying patterns consistent with trying to hide amongst liquidity traders by repurchasing during periods of increased market activity or whether they exhibit patterns consistent with providing liquidity by trading during periods of lower overall activity.

We partition each day into 30 minute intervals and examine repurchase activity within the day. As can be seen in Figure 2, NYSE firms executed over 15% of their repurchases in

¹¹For example, see Chan, Christie, and Schultz (1995) and Jain and Joh (1988).

the first half hour of trading. In fact, 26.3% of all repurchases on the NYSE take place in the first hour of trading for the day. On the Nasdaq, only 15% of repurchases are executed in the opening hour and 25% take place in the final hour of trading. On the NYSE, firms appear to be consistent with other traders in their aggression level throughout the day, taking roughly 5% of the trading volume each half hour. On the Nasdaq however, aggression spikes in the late afternoon where repurchases account for over 40% of trading volume.

Figure 3 presents NYSE and Nasdaq total daily volume for repurchase and all other trades. On the NYSE, trading levels display the familiar u-shaped patterns: volume is higher in the morning and late afternoon than during the middle of the day. One exception to the pattern is the low level of repurchase activity in the closing half-hour of trading. This is consistent with some firms' following part of the SEC's voluntary safe harbor Rule 10b-18.¹² While others have documented u-shaped volume patterns on the Nasdaq, the Nasdaq repurchases in our sample do not appear to exhibit such a pattern. Non-repurchase volume (which includes unidentified repurchases) is roughly constant over the course of the day and increases slightly at the close. Repurchase activity peaks between 3:00 and 3:30 and falls dramatically to only 2% during the final half-hour of trading.

4.1 Determinants of Repurchase Day

The choice of how many shares to repurchase on a given day can be driven by many factors. As can be seen in Figure 1, firms do not trade uniformly over time. We conjecture that the number of shares repurchased should be related to market conditions such as price movement, aggregate trading volume, and market liquidity. We also consider whether firms are timing their trades to take advantage of inside information. If so, we should find trades varying systematically around information releases. We should also see increased buying following price declines and in advance of price increases. In an attempt to determine whether firms are strategically repurchasing, we estimate a model of daily repurchase volume. We include contemporaneous and lagged market condition and firm-specific variables, such as price movement and trading volume as potential motivating factors for a firm to execute a

¹²One provision of 10b-18 precludes repurchasing in the last 30 minutes of the day. Cook, Krigman, and Leach (2000) examine in detail whether 10b-18 affects repurchase behavior and disclosures regarding repurchase behavior.

repurchase. We estimate the following tobit model of the daily repurchase intensity:¹³

$$\begin{aligned} V_{i,t}^R = & \beta_1 + \beta_2 R_t^M + \beta_3 X R_{i,t} + \beta_4 CAR_{i,t-k,t-1} + \beta_5 CAR_{i,t+1,t+k} \\ & + \beta_6 AV_{i,t} + \beta_7 C_{i,t} + \beta_8 A_{i,t} + \beta_9 A_{i,t\pm 2} + \epsilon_{i,t} \end{aligned}$$

The dependent variable is the number of shares repurchased for each firm i on day t ($V_{i,t}^R$). As independent variables, we include the contemporaneous market return proxied by the daily return on the S&P 500 (R_t^M), the excess stock return ($AR_{i,t}$) defined as the stock return less the contemporaneous return on the S&P500, lead and lag cumulative abnormal returns (CAR),¹⁴ abnormal trading volume relative to a non-repurchase environment ($AV_{i,t}$)¹⁵, the percent of the program completed to date ($C_{i,t}$), an indicator variable set to one if the firm made any news announcement on the day ($A_{i,t}$), or in the two days leading up to or following the announcement ($A_{i,t\pm 2}$).

We estimate the model separately for each firm during the complete time series of trade days starting 10 trade days prior to the first reported repurchase through 10 trade days following the last reported repurchase. We report the mean and median estimated coefficients across estimations and the p-values related to a t-test and signed rank test against the null hypothesis that the mean (median) coefficient is equal to zero. We report the results for the full sample of firms and partitioned by exchange listing. Table 4 contains the results.

For the full sample of firms, we find that firms increase repurchasing during periods of high trading volume. Importantly, all firms in our sample curtail repurchasing in the five-day window centered around firm-specific information releases. Thus, they take care to avoid trading on inside information which is going to be publicly disseminated. The results regarding the return-related variables differ by exchange listing. For the 23 NYSE firms in our sample, we find that they repurchase more when the market is contemporaneously down,

¹³We employ tobit estimations because we are modeling a continuous variable, daily repurchase shares, which is censored at zero.

¹⁴We measure cumulative abnormal returns as the buy and hold excess return of the stock relative to the S&P 500 over three-day and five-day windows, in advance of and following days of successful repurchases. We also performed the same estimations using a ten-day CAR window, the results are qualitatively the same.

¹⁵Abnormal Volume is defined as the ratio of daily aggregate volume to normal trading volume measured in the four calendar weeks prior to the week of the announcement of the repurchase program. Thus, the denominator measures ‘normal’ volume which is uncontaminated by the repurchase program.

the excess return of the stock is lower, and when the cumulative abnormal return for the prior 3 and 5 days is more negative. Thus, these firms appear to be price sensitive in the daily repurchase decision. For the 40 Nasdaq firms, we find they repurchase more following periods of negative cumulative abnormal returns. However, they appear to be insensitive to contemporaneous market and own price movements. Neither subsample appears to consistently repurchase in advance of price increases. Overall, the tobit analysis indicates that the NYSE firms have better timing ability, or are more locally price sensitive when repurchasing than are Nasdaq firms.

4.2 Within-Day Repurchase Timing

Barclay and Smith (1988) hypothesize that, by placing limit orders at which it is willing to buy shares, a firm could establish a lower bound on the bid price, and thus help to support a falling stock price. They termed this use of limit orders to buy at the bid the ‘Competing Market-Maker Hypothesis.’ Our regression results indicate that NYSE firms buy on down market days, consistent with both repurchasing as cheaply as possible and posting a stabilizing bid to help support a falling stock price. We examine the timing of trades and trade execution location relative to the bid-ask spread to determine whether corporations are using limit or working orders at the bid in an attempt to establish a floor price.

In order to examine trade execution location and within-day repurchase timing, transaction-level repurchase data are necessary. Most firms in our sample provided total shares repurchased and an average price for each trading day. Some firms provided more detail such as daily volume repurchased at each trading price. We use combinatorial programming across the entire sample to identify repurchase transactions from the *NYSE Transactions and Quotations* (TAQ) database. We provide details of the combinatorial procedure in Appendix B and the detailed results in Appendix C. Overall, of the 3,585 reported repurchases across 64 firms and 2,190 firm days, we locate 2,655 repurchase trades on 1,431 firm days representing over 31 million shares. This is the sample used in our analysis of within-day timing.

As can be seen in Table 5, Panel A, 737 or 40.7% of NYSE repurchase trades, accounting

for over 8 million shares, were bought at the prevailing bid.¹⁶ For Nasdaq firms, we only find 43 of 752, or 7.6% of the repurchases at the prevailing bid. Thus, trade execution location results for the NYSE firms are consistent with Barclay and Smith’s (1988) ‘Competing Market-Maker Hypothesis.’

We also examine intraday returns surrounding repurchase trading. Results are contained in Panel B of Table 5.¹⁷ Consistent with our regression results, we find that NYSE firms are buying on ‘down-trending’ days. The price of NYSE stocks falls by .66% (p -value = .0001) from the open to the close on repurchase days. We partition each trade day return into two parts; return from the open to the repurchase, and return from the repurchase to the close, to determine if firms are buying in advance of the price drop or following the price drop.

On the NYSE, prices fall by -.61% (p -value = .0001) from the open to the repurchase and only drop by an additional -.05% (p -value = .0568) from the repurchase to the close. Thus, on repurchase days, over 90% of the price movement occurs prior to repurchase trades. We find a falling price immediately prior to repurchase, purchasing at a low price (the bid price), and a relatively stable price following the repurchase. Thus, NYSE firms tend to display evidence of within-day timing ability. In contrast, on the Nasdaq, the stock price is flat on repurchase days. Our Nasdaq firms appear to exhibit no obvious evidence of within-day timing. We conjecture that the lack of such evidence is in part due to the unavailability of limit orders for Nasdaq during the period we study.

To investigate the determinants of the intraday timing of trades, we estimate a tobit model on half-hourly repurchase intensity.¹⁸ We partition each day into thirteen 30-minute trading windows ($t = 1, 13$) and examine repurchase activity as it relates to contemporaneous and lagged intraday market conditions. The tobit model estimated is:

$$V_t^R = \alpha_1 + \alpha_2 V_{0,t-1}^{NR} + \alpha_3 V_t^{NR} + \alpha_4 R_{0,t-1} + \alpha_5 R_{t-1,t} + \alpha_6 s_t + \alpha_7 (s_t - s_{t-1}) + \alpha_8 t + \epsilon_t$$

¹⁶We use the Lee and Ready (1991) algorithm for signing trades.

¹⁷We calculate intraday returns using the midpoint of the current bid-ask spread to eliminate the effect of bid-ask bounce in our analysis.

¹⁸We only include days of successful repurchase in the intraday repurchase intensity estimations.

The dependent variable in our estimation is identified repurchase share volume in the t th 30-minute window (V_t^R). The independent variables include non-repurchase trading volume from the market open to time $t - 1$ ($V_{0,t-1}^{NR}$) and during the trading window t (V_t^{NR}), the return on the stock from market open through time $t - 1$ ($R_{0,t-1}$) and from period $t - 1$ to t ($R_{t-1,t}$), the current bid-ask spread (s_t), the change in spread from time $t - 1$ to time t ($s_t - s_{t-1}$), and time index (t), capturing the within day timing of trades.¹⁹

Details of the intraday estimations are contained in Table 6. The results suggest that NYSE firms are much more responsive to variations in intraday market conditions than are Nasdaq firms. On the NYSE, firms repurchase significantly more shares during a trading window when (1) the intraday return is lower (they buy into falling prices), (2) when the bid-ask spread is narrower, and (3) when non-repurchase volume is higher. Additionally, they repurchase less later in the day. On the Nasdaq, repurchase volume is higher when non-repurchase volume is higher and they appear to purchase more later in the day. Overall, the results of the intraday repurchase intensity estimations support the notion that NYSE firms react quickly to, and take advantage of, changing market conditions. On the Nasdaq, there is no similar evidence.

5 Market Timing Ability

We have demonstrated that firms respond to market conditions and information releases when executing transactions. The question remains whether firms can time the market. That is, do they use private information advantageously allowing them to make trading profits on the repurchase portfolio. In this section, we address whether open market repurchase programs are profitable. Motives and objectives regarding the announcement and execution of a repurchase program may vary substantially across firms. Therefore, we base our analysis on the assumption that, *ceteris paribus*, repurchasing firms prefer low acquisition costs and high returns on the repurchase portfolio.

¹⁹Returns are calculated using the midpoint of the bid-ask spread to avoid introducing bid-ask bounce effects into the estimations.

We begin by evaluating the firms' timing decisions on the chosen trading days. That is, was their purchase price above or below the price of other trades on the same day? We compare the share-weighted average repurchase price for each day with the share-weighted average price of all nonrepurchase volume for the day.²⁰ Table 7 Panel A provides details. On the NYSE, firms pay an average of 2 cents less per share than non-corporate trades. The difference is significant at conventional levels. On the Nasdaq, corporations pay nearly 10 cents more per share than non-corporate trades. The difference is statistically significant and economically meaningful representing nearly 1% of the purchase price. Thus, taking the choice of trade day as given, NYSE firms appear to have good execution skills, whereas Nasdaq firms pay significantly higher relative prices.

We next evaluate each firm's skill in selecting days to trade over the course of a repurchase program. The measurement of timing in a repurchase framework is not the same as that of a typical portfolio manager. Traditional measures such as those discussed in Grinblatt and Titman (1993), exploit the ability of traders to buy *and* sell securities. Since U.S. corporations only have discretion over when they buy shares, they face an optimal accumulation problem rather than a traditional market timing problem. Thus, we evaluate actual accumulation strategies against other accumulation strategies that achieve the same ultimate share holdings (and have the same terminal wealth).

The first benchmark we consider relates to the cost of acquiring the same portfolio through different repurchase patterns. We begin with the actual dollar cost of a program:²¹

$$AC_i = \sum_{t \in RD_i} \bar{P}_{it} V_{it}^R \quad (1)$$

where AC_i is the dollar cost for firm i , RD_i is the set of days firm i was successful in repurchasing, $\bar{P}_{it}$ is the average price paid by firm i on day t , and V_{it}^R is the volume (number of shares) repurchased by firm i on day t . The 64 repurchase programs cost an average

²⁰This execution skill measure is similar to that proposed by Manaster and Mann (1996) for assessing execution skill for futures traders, although their time interval is much shorter and they construct benchmarks for buys and sells separately. Benchmarking repurchase execution only against signed buys could be misleading because successful limit order repurchases would usually be signed as sells.

²¹Transactions costs do not include commissions.

of \$45.7 million, with a range of \$147,875 to \$500,000,000. We compare each program’s performance against several, although certainly not all, reasonable benchmarks.

The minimum (maximum) cost strategy assumes that shares were repurchased at the lowest (highest) feasible aggregate price during the repurchase period.²² Panel B of Table 9 contains details. The average NYSE program costs \$116.3 million with a median of \$62.8 million. The NYSE firms pay on average \$12.3 million above the minimum cost, but about \$15 million less than the maximum cost. In percent terms, the median difference from the cost benchmarks is +6.5% and -10.8%. The statistical tests for dollar and percent differences (where each firm provides one observation) indicate that NYSE repurchasing firms do significantly better than the maximum cost benchmark and worse than the minimum cost. The Nasdaq programs in our sample, averaging only \$6.1 million, are significantly smaller than those for the NYSE. Our Nasdaq firms pay on average \$570,652 more than minimum cost, but \$475,490 less than maximum cost. The overall worst execution firm pays nearly \$85 million more than the minimum cost (in a total program value of \$500 million); the best execution is equal to the minimum cost.²³

A more realistic benchmark might be the cost of buying the same number of shares by accumulating them uniformly over the entire reported repurchase period.²⁴ For each day, starting with the first reported repurchase through the last reported repurchase, our “uniform” strategy buys a fixed number of shares every day at the closing price.²⁵ The uniform repurchase cost (UC) is defined as:

$$UC_i = \frac{\sum_{t=1}^{T_i} P_{it}^c}{T_i} V_i^R \quad (2)$$

where T_i is the number of trading days in the repurchase horizon for firm i , $V_i^R = \sum_t V_{it}^R$ is

²²These measures represent theoretical upper and lower bound costs to acquire the reported repurchased shares during each repurchase program.

²³One firm repurchased on only one day and traded at the lowest price for the day.

²⁴We begin the accumulation on the day of the first actual repurchase which assures that some funds were available to consummate repurchases. Prior to this date an announcement of a program does not assure that the capital has been made available to those responsible for repurchase execution.

²⁵We do not claim that firms could execute our benchmark strategies. For example, SEC 10b-18 provisions restrict firms conforming to 10b-18 from repurchasing during the last half-hour of the trading session.

the number of shares in the terminal repurchase portfolio, and P_{it}^c is the closing price on day t for firm i .

A second naive benchmark buys in proportion to each day's trading volume. The proportional repurchase cost (PC) for firm i is defined as:

$$PC_i = \frac{\sum_{t=1}^{T_i} V_{it} P_{it}^c}{V_i} V_i^R \quad (3)$$

where $V_i = \sum_t V_{it}$ is the total volume over the repurchase horizon and V_{it} is the total shares traded on day t in stock i . The third naive strategy we consider is repurchasing 'as soon as possible' (ASAP) subject to trading volume for the day. We assume the corporation repurchases 100% of the trading volume beginning the first day of an actual repurchase and continues taking 100% of each consecutive day's volume until all desired shares are repurchased. The final benchmark we consider examines the cost of buying the program at the closing price for each actual repurchase day. Thus, this measure examines, from a program-wide perspective, the within-day timing of trades. The closing repurchase cost (CC) for firm i is defined as:

$$CC_i = \sum_{t \in RD_i} P_{it}^c V_{it}^R \quad (4)$$

where CC_i is the dollar closing cost for firm i , RD_i is the set of days firm i was successful in repurchasing, P_{it}^c is the closing price for firm i on day t , and V_{it}^R is the volume (number of shares) repurchased by firm i on day t .

Panel B of Table 7 provides details for these additional cost-based benchmarks. Apparent differences between NYSE and Nasdaq timing skill emerge. We find that the 41 Nasdaq programs pay more than if they had repurchased as soon as possible. That is, their decision to wait to repurchase was on average \$319,048 more expensive than if they had repurchased all shares as quickly as possible. The average percent difference from the ASAP benchmark is 1.3% with 63% of Nasdaq firms' having higher than ASAP benchmark costs. Nasdaq costs were not significantly different from the uniform or proportional benchmarks. Relative to the closing cost benchmark, the Nasdaq firms paid significantly more in both dollars

and percentage terms. Overall, 30 of the 41 Nasdaq firms underperformed this simplistic benchmark.

The 23 NYSE firms, in contrast, appear to exhibit some timing skill. Their acquisition costs are lower than the uniform and proportional benchmarks, saving an average of \$1.4 million per program. Only 17.4% of the NYSE firms paid more than the uniform benchmark cost and 21.7% paid more than the proportional benchmark cost. While NYSE firms' acquisition costs were not significantly less than the ASAP costs, 14 of 23 firms (60.8%) did pay less than purchasing as soon as possible.

Figure 4 provides a graphical representation of the actual repurchase cost as a percent spread between actual costs and the benchmark costs (minimum, maximum, ASAP, uniform and proportional) for each program on the Nasdaq and the NYSE separately. As can be seen, the averages mask the fact that some Nasdaq firms have good cost-based performance and a few NYSE firms have poor cost-based performance. Examination of the proportional benchmark reveals that 5 of 23 NYSE programs had timing decisions worse than a proportional approach, and 12 of 41 Nasdaq firms actually beat a proportional strategy. The non-uniformity of results within exchange-listing is consistent with the diversity of styles we highlighted in Section 3. Perhaps most provocative is the suggestion that NYSE firms' accumulation strategies appear on average to outperform those of Nasdaq firms.²⁶

6 Liquidity Impacts

The final issue we address is whether the presence of the corporation in the market affects the liquidity of the stock. There are two ways that a firm, through a repurchase program, can affect the liquidity of a stock. First, Barclay and Smith (1988) suggest that regular repurchases can provide competition for market makers and potentially make a firm's shares

²⁶Risk and time adjusted excess wealth creation measures provide no additional insight. We have calculated expected excess wealth based on (1) the ex ante risk-free rate and (2) a CAPM-based rate of return, and find no evidence of significant abnormal wealth creation due to repurchase execution. However, consistent with the cost-based analysis, NYSE firms do beat the naive uniform and proportional repurchase strategies even after controlling for risk.

more liquid. By posting limit orders to buy, the firm can establish a lower bound price for a stock and in doing so, effect a narrowing of the bid-ask spread. In this case, increased liquidity is defined as a narrowing of the bid-ask spread resulting in lower trading costs. Alternatively, Barclay and Smith (1988) argue that an asymmetric-information-effect of possibly trading with an informed trader (the corporation) could dominate any market-making-effect and result in a widening of the bid-ask spread.²⁷ Even if there is a wider spread during the repurchase period overall, our dataset allows us to examine temporary narrowing in this wider spread around the time the corporation is buying to see if firms are competing with other buyers to provide liquidity to sellers.

A second means of affecting market liquidity is to absorb sell-side pressure by standing ready to buy shares at the bid without allowing the price to fall. To examine this aspect of liquidity, we consider the change in a firm’s stock price in the presence of sell-side pressure.²⁸ The higher the change in price resulting from a trade, the lower the liquidity. Following Breen, Hodrick, and Korajczyk (1999) we examine whether the price impact of trades differs on repurchase days from that on non-repurchase days. That is, we examine if repurchasing firms, by absorbing sell-side order imbalances, limit the negative price impact generally associated with large order imbalances.

6.1 Bid-Ask Spread Effects

Isolating repurchase impact on spreads is challenging because we do not know: (i) how and when managers communicate or signal their intent or willingness to buy to a broker; and, (ii) how brokers use, or communicate to others, corporate intent.²⁹ Communication may not

²⁷Others investigating the potential spread widening in the United States have not always concurred that there is a repurchase-related widening in the spread. See e.g., Miller and McConnell (1995) and Singh, Zaman, and Krishnamurti (1994). Brockman and Chung (2000) investigate the liquidity impact of repurchasing in Hong Kong. They document spread widening on repurchase days, consistent with an asymmetric information effect. Importantly, repurchase trading is disclosed by the start of trading on the following business day in Hong Kong. Thus, when a repurchase program is initiated (as opposed to announced), market participants are informed of the corporate presence.

²⁸Since firms can only buy shares, they can only confer this type of liquidity benefit during periods of sell-side order imbalance.

²⁹It is also important to consider the impact of alternative market structures in this type of analysis. The limit order book on the NYSE provides a direct venue for firms to influence the spread proactively. The

constitute an explicit order. For instance, the repurchasing firm may call a broker and seek current quotes while indicating that it may be in the market for 50,000 shares. While not an order, the communication can influence quotes.

As can be seen in Figure 5, for the NYSE, we find a small spread widening relative to a pre-repurchase period.³⁰ The mean bid-ask spread widens by 1.3 cents (significant at the .0001 level). On the Nasdaq, however, the bid-ask spread is significantly narrower on repurchase days than during the pre-repurchase period. The mean pre-repurchase spread is \$0.548 compared to \$0.362 on repurchase days. Thus there is no evidence of an asymmetric-information-related widening in Nasdaq spreads.

To address liquidity provision within the repurchase period, we make paired comparisons of spreads on repurchase days to those on adjacent non-repurchase days.³¹ We use a transactions-based spread calculation methodology to reflect transaction-relevant spreads. The statistics are computed using the spreads attached to actual transactions. This procedure ignores regional autoquotes, and other types of non-competitive and stale quotes.

We define the ask (bid) prevailing at the time of the n^{th} transaction on firm i 's stock on day t as a_{nt}^i (b_{nt}^i).³² The spread is defined as $s_{nt}^i = a_{nt}^i - b_{nt}^i$ with a daily average spread for firm i on day t of $\bar{s}_t^i$. We focus on comparisons of $\bar{s}_t^i$ for repurchase days and their lead and lag day counterparts. Our statistics are based upon a pooling of firms and trade days, where t is a repurchase day, with abnormal spread (AS) observations defined by: $AS_{it}^{IG} = \bar{s}_t^i - \bar{s}_{t-1}^i$ and $AS_{it}^{LD} = \bar{s}_t^i - \bar{s}_{t+1}^i$.

Table 8 contains daily abnormal spread measures. The abnormal spreads relative to absence of a viable limit order book on the Nasdaq (during the time of this study) makes it difficult for corporations to have the same influence.

³⁰The analysis relative to the pre-repurchase period is conducted for the 50 firms for which we could identify a clean announcement date, uncontaminated by a pre-existing repurchase program.

³¹The pairing process results in 642 days for the full sample of 64 firms and 65 tickers with lead and lag benchmarks. A lead (lag) benchmark refers to the day before (after) a repurchase day when the corporation did not repurchase. Thus, if a firm repurchases on three consecutive days, the first day will have a lead benchmark and the third day will have a lag benchmark. The second day will have neither a lead nor lag benchmark.

³²The prevailing bid and ask refer to the best bid and offer (BBO) which we calculate for NYSE stocks using quotes from all regional exchanges. TAQ reports only the BBO for NASDAQ stocks, not the individual quotes from which the BBO is calculated.

lag and lead benchmarks are significantly negative.³³ The results hold for both the NYSE and the NASDAQ sub-samples. For example, the mean of daily average bid-ask spreads for the 350 lead days in the NASDAQ sub-sample is \$.457. The 350 associated corporate trade days have a mean daily average spread of \$.445. The set of 293 trade days for the NYSE firms have an average spread of \$.204, compared to a lag day average spread of \$.221, representing a $-\$.017$ (significant at a 6% level) abnormal spread. Relative to the day following the corporate trade day, the mean NYSE corporate trade day abnormal spread is $-\$.007$, significant at the 1% level. Overall, our results support the notion that open market repurchase executions can, through the narrowing of spreads, provide a liquidity benefit to the market.

Consistent with our goal of characterizing repurchase execution strategies, our emphasis is primarily the documentation of statistically significant patterns in repurchase behavior and market impact. Nonetheless, we also believe that many of our results provide evidence of the economically significant impact repurchase programs can have on NYSE and Nasdaq trading. For a change in spread of 18.6 cents relative to the pre-repurchase period for Nasdaq firms, there is little doubt that the economic stakes are significant. We would argue that even a \$.02 change in spread (NYSE) for the repurchase day represents a potentially significant economic impact for the large volume of other traders active on a repurchase day.

6.2 Price Impact Liquidity Effects

The second measure of liquidity we consider is the change in a firm's stock price associated with its observed trading volume. The higher the change in price resulting from a trade, the lower the liquidity. Following Breen, Hodrick, and Korajczyk (1999) we examine whether the price impact of trades is different on repurchase days compared to non-repurchase days. We partition each trading day into thirteen 30-minute trading windows and calculate return using the midpoint of the quoted bid-ask spread during each window and net turnover

³³See Ahn, Cao, and Choe (1996) for a microstructure discussion of the economic significance of changes in quoted bid-ask spreads. They demonstrate the economic and statistical significance of small dollar spread changes, e.g. 1 cent.

(*NTO*) defined as buyer-initiated volume less seller-initiated volume as a fraction of shares outstanding.³⁴

We pool all firms and days together and use the half-hourly observations of return and *NTO* to estimate the relationship between net turnover and intraday returns on repurchase and non-repurchase days. We include an indicator variable (*R*) set to 1 if the firm made a repurchase on a day and interact the indicator with the *NTO* variable. We also control for whether *NTO* is positive (buy-pressure) or negative (sell-pressure) to identify the impact of repurchasing behavior under each scenario. We include an indicator variable (*Neg*) to indicate the *NTO* during the period was negative, and an indicator (*Pos*) to indicate if the *NTO* was positive. The estimation equations are:

$$\text{Return} = \alpha_1 + \beta_1 \text{NTO} + \alpha_2 \text{Neg} + \beta_2(\text{Neg} \cdot \text{NTO}) + \alpha_3(\text{R} \cdot \text{Neg}) + \beta_3(\text{R} \cdot \text{Neg} \cdot \text{NTO}) + \epsilon$$

$$\text{Return} = \alpha_1 + \beta_1 \text{NTO} + \alpha_2 \text{Pos} + \beta_2(\text{Pos} \cdot \text{NTO}) + \alpha_3(\text{R} \cdot \text{Pos}) + \beta_3(\text{R} \cdot \text{Pos} \cdot \text{NTO}) + \epsilon$$

Under periods of sell pressure, if the corporation is absorbing sell imbalances through repurchasing, β_3 should be negative. This would indicate that large sell imbalances do not have as large an impact on prices as the corporation is absorbing the imbalance through repurchasing. Since the corporation is only purchasing shares, under periods of buying pressure (positive *NTO*) repurchasing cannot absorb the order imbalance, but should be a contributing factor to it. Thus, under periods of buy pressure, if the firm is an aggressive buyer, we would expect a positive coefficient on β_3 .

Results of the estimations are contained in Table 9. For Nasdaq firms, we find a significantly negative coefficient on β_3 , consistent with liquidity provision during periods of sell pressure. For the NYSE firms, during periods of sell pressure, repurchasing contributes marginally to market liquidity as evidenced by the negative (although not statistically significant) coefficient on β_3 . In periods of buy pressure, both the NYSE and the Nasdaq

³⁴Following Breen, Hodrick, and Korajczyk (1999), we classify every trade as buyer initiated or seller initiated based on whether the trade price is greater than or less than the quote mid-point. We use a 5 second lag when assigning trades to quotes using the TAQ database, consistent with the findings of Odders-White (1998).

estimations yield significantly negative coefficients on β_3 . The results are consistent with repurchasing firms not driving up stock prices during periods of repurchasing. Thus, the approach suggested by Breen, Hodrick, and Korajczyk (1999) for evaluating market liquidity provides some evidence that large buy and sell order imbalances have lower price impact when repurchasing is taking place.

7 Concluding Remarks

It is difficult to characterize a ‘typical’ repurchase program. We find that firms exhibit a diversity of styles in executing actual repurchases. Some appear to be reacting to falling prices and attempting to stem or even reverse the decline. Others appear to exhibit little concern about market impact as they repurchase aggressively. Typically, repurchasing firms do not seem to exploit short-term trading opportunities around public announcements by the firm. Perhaps one of the more interesting findings in this investigation is the apparent interaction between repurchase execution and exchange trading structure. NYSE and NASDAQ repurchase executions appear to exhibit distinct differences in trading style and performance relative to naive benchmarks.

For NYSE firms, we find execution patterns consistent with advantageous market timing. NYSE firms beat several naive accumulation strategies and appear to use repurchases to lean against falling stock prices. Their intensity of repurchasing is sensitive to changing market conditions across and within days. They buy following price drops and in advance of rising prices. For NASDAQ firms, we find no evidence of market timing ability. Nasdaq firms’ timing decisions are no better than the naive strategies we evaluate. Their daily repurchasing levels appear to be related only to trading volume and excess returns on the prior few days. Although they repurchase following days of negative excess returns, they display no evidence of timing ability when aggregated over the entire repurchase program. Consistent with their lack of timing on a daily basis, their intraday repurchasing appears to be unrelated to changing market conditions such as intraday returns and bid-ask spreads.

It appears that the differences in the execution style and performance of NYSE and Nasdaq firms are partly due to the inability of Nasdaq repurchasing firms to use limit orders (in our sample). Perhaps the strongest evidence for this relevance of microstructure-related differences is that NYSE firms succeed in purchasing at the bid far more often than do Nasdaq firms. In addition to microstructure-related differences in our sample firms we also note that our NYSE firms are larger and run larger repurchase programs on average than our Nasdaq firms. Consequently, NYSE firms may have more incentive and resources to devote to advantageous execution strategies than their Nasdaq counterparts. Consistent with this notion, several of the Nasdaq firms in our sample indicated that they delegate the repurchase program to broker(s).

Open market repurchase programs are an increasingly important aspect of a company's payout policy and its treasury function. Due to the veil behind which repurchase trading takes place, we know very little about how open market repurchase programs are executed. While our initial study indicates that treasurers may need to consider their stock's trading environment and the impact it has on naive measures of program success, major differences remain in execution style that are not due merely to microstructural differences. As our earlier graphs and summary statistics indicate, repurchase programs differ tremendously in lags between announcement and execution, trading intensity, length of program and sensitivity to own price and market conditions. Although it is not obvious how more complete and comprehensive U.S. data will be forthcoming as long as publicly traded corporations need not disclose execution data, our initial study of voluntarily disclosed NYSE and Nasdaq data suggests exchange-related and other differences that are deserving of further scrutiny.

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Appendix A

OPEN MARKET STOCK REPURCHASE SURVEY AND DATA SHEET

Firm Name:

Type of Security Repurchased (e.g. Leach Corp. Class A Common):

Targeted Program

Starting Date:

Dollar Amount:

Number of Shares:

Program Status

___ Completed

___ Ongoing

___ Suspended or withdrawn

___ Uncertain

If a target number or value of shares was not achieved, why was the program suspended or withdrawn?

Position title or type of agent designated to coordinate repurchase trading:

Is there someone we can contact if we have questions?

Would you like to receive a copy of the completed study report (completed form is required)?

General comments on trade execution and costs:

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<u>Date</u>	<u>Exchange</u> (If <u>Known</u>)	<u>Number of Shares</u> (<u>Actual not Round Lots</u>)	<u>Price Paid per Share</u> (<u>1/8ths or Decimal</u>)	<u>Price Aggregation*</u> (<u>E= Exact, A=Average</u>)	<u>Comments</u>
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Appendix B

To examine repurchase execution skill, transaction-level repurchase data are necessary. Most firms in our sample provided aggregate volume and average price for a given trading day. Some firms provided daily volume repurchased at a given trading price. For low trading volume stocks, visual identification of repurchase transactions in the TAQ data is straightforward. Consider the following example. On one day, a firm reported repurchasing 31,000 shares at a price of \$7.125 per share. The following is the TAQ listing of transactions in that stock for the given day:³⁵

<u>Print</u>	<u>Time</u>	<u>Bid</u>	<u>Asked</u>	<u>Price</u>	<u>Shares</u>
1	34268	\$6.750	\$7.500	\$7.500	100
2	35841	6.750	7.500	7.500	500
3	42677	6.750	7.500	7.000	31000
4	42700	6.750	7.500	7.125	31000
5	45273	6.750	7.500	6.750	500
6	51974	6.750	7.500	7.500	100
7	54702	6.750	7.500	7.500	200
8	57105	6.750	7.500	7.125	1000
9	58163	6.750	7.500	7.000	300

Transaction #4 is the only feasible execution of 31,000 shares at a price of \$7.125. No grouping of other trades provides the required price-volume combination. In other cases, the reported volume is not feasibly executed as a single print. For example, another corporation reported repurchasing 26,500 shares at an aggregate cost (exclusive of commissions) of \$542,437.50. There were five transactions in the stock for that day:

<u>Print</u>	<u>Time</u>	<u>Bid</u>	<u>Asked</u>	<u>Price</u>	<u>Shares</u>
1	41293	\$20.250	\$20.750	\$20.250	20000
2	41349	20.250	20.750	20.500	20000
3	41771	20.250	20.750	20.250	1200
4	42879	20.250	20.750	20.250	4000
5	42907	20.250	20.750	20.375	6500

The only feasible combination of trades that achieves the reported cost-volume repurchase is #2 and #5. For both of these low volume examples, perfect identification of reported repurchases is not difficult.

For high volume stocks, visual identification is not feasible and computer-based searching is required. We use combinatorial methods to identify combinations of a day's prints that together equal the reported repurchase price and volume. Due to computing constraints, we stop searching on high volume days at quadruplets.³⁶ As we are unable to search through all feasible combinations of prints, we consider simplicity. If a repurchase can be explained by a

³⁵Time is presented in seconds from midnight. The markets open at 33,480 seconds, or 9:30 a.m.

³⁶For one NYSE firm, we could only search through combinations of three prints. For this firm, on one day there were 2,951 transactions in the stock; a search for all combinations of four prints requires examining 3,153,423,033,475 combinations.

single print, we ‘identify’ the print as a repurchase. We discontinue our search for that volume and price at higher complexity levels, e.g. in two prints (pairs). If a reported repurchase cannot be explained by a single print, but is consistent with a unique pair, we treat the pair as repurchases and discontinue our search. If a paired execution is not feasible, we consider triplets and so on. Using this heuristic approach, we can ‘identify’ 1,981 repurchases: 1,114 singletons, 228 pairs, 85 triplets and 39 quadruplets across 63 tickers.

As might be anticipated, the combinatorial search frequently generates more than one matching combination at a given level of complexity (e.g., 15 pairs). In these instances, we search for common trades occurring in all combinations at the given level of complexity. For example, if we are searching for 10,100 shares at \$30.00, and find 15 feasible pairs, each containing the same 10,000 share at \$30.00 print but matched with 15 different 100 share at \$30.00 prints, we classify the 10,000 share print as an identified repurchase in a partially identified pair. The 100 share portion remains ‘unidentified.’ We can identify an additional 674 prints using this within-complexity procedure. Overall, of the 3,585 reported repurchases across 2,190 firm days, we can identify 2,655 prints on 1,431 firm days representing over 31 million shares.

We recognize that our heuristic is not perfect and offer three defenses:

1. The measures of execution timing and skill that consider aggregate daily reported repurchase price and volume are not subject to error from transaction misidentification.
2. Misidentification will tend to diminish differences and bias against significance in our transactions-level tests.
3. We ran exhaustive combinations on the subset of reported repurchases where (i) there were not more than 30 prints in the TAQ data for the day; or (ii) there were not more than 100 TAQ prints and not more than 3,000 shares repurchased for the day.³⁷ This exhaustive searching identifies 504 trades. These must be repurchases if the firms provided accurate disclosure. All transaction-specific tests are replicated on this smaller exhaustive subset; while the power of the tests decline, all results are qualitatively the same.

³⁷The 3,000 share requirement limits the search to combinations of not more than 30, as prints are reported on TAQ in round lots of 100 shares. The choice of a maximum complexity level of 30 for the exhaustive search is dictated by computing constraints.

Appendix C

Details of corporate repurchase transaction identification are presented. Combinatorial programming, searching for up to combinations of four transactions identified 2,655 corporate repurchase executions from the 3,585 reported to us by corporations. Summary measures include the number of identified prints, the average number of shares per identified transaction, and the percent of partially identified transactions which we identified.

	Total		NYSE		NASDAQ	
	Mean	Median	Mean	Median	Mean	Median
Total Identification						
Reported Transactions		3,585		2,880		705
Reported Shares	81,215,500		67,045,500		14,170,000	
Prints Identified		2,655		2,023		632
Shares Identified	31,003,100		19,829,600		11,173,500	
Singletons						
Full Identification						
Singletons		1114		603		511
Shares	14,366	7,000	12,384	5,000	16,705	10,000
Pairs						
Full Identification						
Pairs		228		190		38
Prints		456		380		76
Shares	21,442	10,000	16,633	6,850	45,486	31,250
Partial Identification						
<i>1 of 2 Transactions Common Across Non-Unique Pair</i>						
Prints		176		168		8
Shares	6,698	3,000	6,255	2,700	16,000	9,500
Percent	68.2%	75.0%	68.7%	76.0%	57.5%	55.1%
Triplets						
Full Identification						
Triplets		85		82		3
Prints		255		246		9
Shares	21,730	12,000	21,684	12,000	23,000	28,000
Partial Identification						
<i>1 of 3 Transactions Common Across Non-Unique Triplet</i>						
Prints		121		117		4
Shares	11,758	5,300	11,450	5,000	25,750	20,500
Percent	60.4%	62.5%	60.4%	62.5%	61.3%	60.9%
<i>2 of 3 Transactions Common Across Non-Unique Triplet</i>						
Prints		155		147		8
Shares	23,719	10,450	24,247	11,400	14,225	6,500
Percent	84.2%	88.7%	84.9%	89.9%	71.8%	74.8%
<i>3 of 3 Transactions Common Across Non-Unique Triplet*</i>						
Prints		3		3		
Shares	10,000	10,000	10,000	10,000		
Percent	100.0%		100.0%			
Quadruplets						
Full Identification						
Quadruplets		39		37		2
Prints		156		148		8
Shares	36,712	19,000	26,318	18,500	229,000	229,000
Partial Identification						
<i>1 of 4 Transactions Common Across Non-Unique Quadruplet</i>						
Prints		59		58		1
Shares	12,971	6,700	12,332	5,900	50,000	50,000
Percent	53.0%	51.2%	53.2%	51.9%	41.7%	41.7%
<i>2 of 4 Transactions Common Across Non-Unique Quadruplet</i>						
Prints		70		66		4
Shares	27,977	14,000	28,657	14,000	16,750	16,750
Percent	79.8%	88.1%	76.9%	88.1%	78.3%	78.3%
<i>3 of 4 Transactions Common Across Non-Unique Quadruplet</i>						
Prints		90		87		3
Shares	21,076	9,900	21,462	9,900	9,900	9,900
Percent	90.8%	95.2%	90.5%	94.7%	99.0%	99.0%

* Two identical prints (same time, price, volume) in the TAQ data resulted in two ways to identify a 10,000 share corporate repurchase using three prints. Thus, the corporate volume was fully identified.

Table 1: Summary Data

Summary statistics for the 64 firms executing repurchase programs including firm size and trading activity, size of repurchase program, length of time to complete, and time from announcement to the first corporate trade are presented.

	Mean	Median	Minimum	Maximum
Days of Successful Repurchase up to Response Date	33.7	15	1	288
Market Capitalization (\$ 000) Year End 1993	2,610,067	272,927	6,111	31,923,201
Trading Volume (000) Shares Cumulative 1993	3,256,412	1,287,343	8,182	24,058,200
Program Shares Repurchased up to Response Date	1,288,830	433,350	18,400	12,372,500
Repurchase Shares as % of Daily Volume	23.28%	18.20%	0.03%	100.00%
Target Shares As % of Shares Outstanding	7.07%	5.11%	0.56%	32.80%
Reported Shares Repurchased to Response Date as % of Shares Outstanding	3.67%	2.91%	0.04%	15.22%
Days from Announcement to 1st Reported Trade*	17	9	1	124
*Calculated for the 50 firms for which an announcement was available and for which there was no evidence of an existing repurchase program.				

Table 2: Survey Response Statistics

Summary statistics are provided for the 64 firms which complied with our solicitation for data and for 390 of 414 firms which chose to ignore our solicitation for which we could locate data. The table contains information on exchange listing, firm size, annual trading volume and announced repurchase program size as a percent of shares outstanding. We also provide the cumulative abnormal return over the announcement period days 0 to +1, and a size-adjusted and value-weighted excess return for the three years post announcement. Parametric and non-parametric tests are provided which compare responding to non-responding firms.

	Responding Firms		Non-Responding Firms			
	Number	Percent	Number	Percent		
Number of Firms	64		390			
Exchange Listing:						
NYSE	24	37.5%	174	44.6%		
Nasdaq	41	64.1%	193	49.5%		
AMEX	0	0.0%	23	5.9%		
	Responding Firms		Non-Responding Firms		t-statistic	Ranked Sums
	Mean	Median	Mean	Median		
Market Value of Equity (\$ 000)	2,646,276	263,475	781,280	261,055	2.29 (.0256)	0.12 (.9014)
1993 Annual Trading Volume (million shares)	3,256	1,287	38,576	12,377	-8.78 (.0001)	-9.05 (.0001)
Targeted Shares as a % of Shares Outstanding	6.90%	5.01%	6.97%	5.08%	0.08 (.9369)	-0.18 (.8534)
CAR at Announcement	2.94%	2.35%	2.23%	1.85%	0.73 (.4664)	2.04 (.7777)
3-Year Size Adjusted Return	39.57%	29.48%	40.29%	29.11%	-0.05 (.9574)	-3.21 (.6550)
3-Year Market Adjusted Return	18.07%	4.71%	19.31%	5.98%	-0.11 (.9112)	2.45 (.7333)

Table 3: Daily Repurchase Activity

The percent of daily volume taken by the corporation is presented for the 2,190 repurchase days in the sample. Data are partitioned by exchange listing. Mean and median values are provided

	NYSE Firms		Nasdaq Firms	
	Mean	Median	Mean	Median
Program Length	195.0	205.0	116.0	85.0
Repurchase Days	66.0	50.0	16.4	11.0
% of Aggregate Volume	6.6%	3.9%	10.4%	7.9%

	NYSE Firms			Nasdaq Firms		
	Days	Mean	Median	Days	Mean	Median
All Days	1577	19.48%	13.50%	613	33.03%	31.20%
Monday	295	19.69%	14.74%	116	33.14%	32.51%
Tuesday	328	19.25%	12.97%	120	33.41%	31.67%
Wednesday	328	20.15%	13.50%	114	33.04%	31.59%
Thursday	319	19.55%	13.92%	131	33.39%	28.38%
Friday	307	18.75%	11.95%	132	32.24%	31.89%

Table 4: Tobit Estimation of Daily Repurchase Intensity

Daily repurchase volume is estimated in a tobit regression for each firm. The dependent variable is the daily number of shares repurchased. Independent variables include contemporaneous, lagged and lead values of the return on the market, excess returns relative to the market, the percent of the repurchase program completed, relative volume, defined as each day's aggregate trading volume divided by the average trading volume during a non-repurchase horizon for each stock, and indicator variables for days when the firm made public announcements. All volume variables are scaled by 100,000. We present the mean and median coefficient from the individual firm estimations and a parametric and non-parametric p-value against the null hypothesis that the mean (median) coefficient is equal to zero.

	Full Sample		Full Sample		NYSE		NYSE		Nasdaq		Nasdaq	
	Mean	Median	Mean	Median	Mean	Median	Mean	Median	Mean	Median	Mean	Median
Intercept	-1.025 (.0001)	-0.347 (.0001)	-0.879 (.0001)	-0.397 (.0001)	-1.816 (.0080)	-0.678 (.0001)	-1.301 (.0004)	-0.732 (.0001)	-0.637 (.0001)	-0.347 (.0001)	-0.637 (.0001)	-0.333 (.0001)
S&P 500 Return	-3.518 (.1306)	-1.900 (.0430)	-4.379 (.0905)	-1.568 (.0704)	-10.860 (.0376)	-4.770 (.0058)	-11.378 (.0300)	-3.276 (.0024)	-2.194 (.6049)	-0.570 (.8846)	-0.356 (.8960)	-0.273 (.9055)
Excess Return	-4.038 (.1518)	-0.379 (.0156)	-3.116 (.0530)	-1.028 (.0010)	-9.689 (.2070)	-2.211 (.0114)	-4.878 (.1688)	-2.920 (.0058)	-1.433 (.1416)	-0.196 (.1270)	-2.103 (.1793)	-0.511 (.0668)
3 Day Car Lag	-3.212 (.1639)	-0.580 (.0015)			-7.777 (.2158)	-1.429 (.0249)			-0.399 (.3920)	-0.216 (.2818)		
3 Day Car Lead	-3.395 (.3402)	0.034 (.8013)			-8.842 (.3708)	0.288 (.0931)			-0.092 (.8737)	-0.146 (.2641)		
5 Day Car Lag			-2.236 (.1059)	-0.793 (.0001)			-3.227 (.3278)	-1.989 (.0126)			-1.667 (.1449)	-0.574 (.0028)
5 Day Car Lead			-0.899 (.3516)	-0.104 (.4641)			-1.448 (.5681)	-0.036 (.9765)			-0.584 (.2693)	-0.203 (.3066)
Relative Volume	0.185 (.0001)	0.105 (.0001)	0.210 (.0001)	0.101 (.0001)	0.254 (.0351)	0.101 (.0021)	0.268 (.0143)	0.112 (.0001)	0.179 (.0001)	0.092 (.0001)	0.177 (.0002)	0.090 (.0001)
Percent of Program Completed	0.778 (.0184)	0.220 (.0001)	0.515 (.0001)	0.255 (.0001)	1.721 (.0527)	0.357 (.0001)	1.012 (.0111)	0.361 (.0001)	0.086 (.3730)	0.115 (.0173)	0.229 (.0077)	0.180 (.0007)
Announcement	-1.570 (.0001)	-0.600 (.0001)	-1.488 (.0001)	-0.633 (.0001)	-2.028 (.0237)	-0.241 (.0188)	-1.742 (.0382)	-0.253 (.0188)	-1.237 (.0001)	-0.713 (.0001)	-1.342 (.0001)	-0.698 (.0001)
Announcement w/in 2 days	-1.469 (.0001)	-0.296 (.0001)	-1.384 (.0001)	-0.291 (.0001)	-1.801 (.0471)	-0.172 (.0092)	-1.508 (.0721)	-0.174 (.0155)	-1.155 (.0001)	-0.493 (.0001)	-1.313 (.0001)	-0.580 (.0001)
Observations		63		63		23		23		40		40

Table 5: Trade Execution Location and Intraday Returns

Panel A contains the signed trade execution location of all repurchase transactions partitioned by exchange listing. We use the Lee and Ready (1991) algorithm for signing trades. We provide the number or trades, the share volume and the percent of the repurchase program that was executed below the midpoint of the bid ask spread (sell), above the midpoint of the spread (buy), trad which we were unable to categorize (unsigned), and those that were executed at the prevailing market bid. Panel B examines intraday returns on repurchase days. We partition each trade day into two partitions; from the open to the repurchase trade, and from the repurchase trade to the market close.

NYSE Repurchases					Nasdaq Repurchases				
Panel A: Signed Execution Location									
	Sell	Buy	Unsigned	At Bid	Sell	Buy	Unsigned	At Bid	
# of Trades	860	786	377	737	159	398	195	43	
Share Volume	8,883,400	7,529,800	3,416,400	8,077,800	2,306,600	6,995,300	3,871,200	995,700	
% of Program	44.8%	38.0%	17.2%	40.7%	17.5%	53.1%	29.4%	7.6%	
Panel B: Returns from Quote Midpoint to Quote Midpoint:									
	Mean	Median	t-stat	p-value	Mean	Median	t-stat	p-value	
Open to Close	-0.66%	-0.23%	-12.01	(.0001)	-0.04%	0.00%	-0.70	(.4873)	
Open to Repurchase	-0.61%	-0.14%	-12.32	(.0001)	-0.02%	0.00%	-0.39	(.6946)	
Repurchase to Close	-0.05%	0.00%	-1.91	(.0568)	-0.02%	0.00%	-0.71	(.4788)	

Table 6: Tobit Estimation of Intradaily Repurchase Intensity

Each day is partitioned into 13 one-half hour trading windows. The dependent variable in the estimation is the number of shares repurchased in each time window. Independent variables include contemporaneous and lagged values of returns, trading volume, bid-ask spreads, and the time index (1 to 13 for the trading window).

	NYSE		Nasdaq	
	Estimate	p-value	Estimate	p-value
Intercept	-0.795	(.0001)	-1.663	(.0001)
Non-Repurchase Shares (t)	1.78E-06	(.0001)	2.70E-05	(.0001)
Return from Open to (t)	-1.649	(.0223)	1.665	(.4078)
Return from (t-1) to (t)	-7.621	(.0070)	10.059	(.1167)
Bid-Ask Spread	-0.684	(.0004)	-0.080	(.2841)
Change in Bid-Ask Spread	0.051	(.0709)	-0.411	(.3920)
Time Index	-0.040	(.0001)	0.031	(.0001)
Cumulative Non-Repurchase Shares	8.89E-08	(.0576)	-2.22E-06	(.0001)
Log Likelihood	-22742.5		-8276.1	
N	11,986		6,617	

Table 7: Repurchase Price and Cost Comparisons

Actual repurchase cost is compared to several repurchase strategies over the repurchase horizon. Panel A contains the daily repurchase price paid relative to the non-corporate trade price on the day of the repurchase presented in dollars and as a percent of weighted average daily price for the 2,190 repurchase days in the sample. Mean and median values are provided. P-values from paired t-tests against the null hypothesis of no difference in price paid (dollar and percent) by the corporation relative to the market are also provided. Panel B contains mean and median values for the 64 repurchase programs in the sample partitioned by exchange listing. Panel B provides a comparison of actual repurchase program cost to the benchmarks based on dollar and percentage differences. p-values corresponding to tests of the null hypothesis that the mean and median are equal to zero are provided in parentheses.

Panel A		NYSE		Nasdaq	
		Price Relatives		Price Relatives	
		Dollars	Percent	Dollars	Percent
	Mean	\$ (0.0202)	-0.056%	\$ 0.0998	0.865%
	Median	\$ (0.0063)	-0.018%	\$ 0.0948	0.669%
	t-stat	-3.58	-3.83	13.47	13.46
	p-value	(.0004)	(.0001)	(.0001)	(.0001)

Panel B		Full Sample		NYSE		Nasdaq	
		Cost Relatives		Cost Relatives		Cost Relatives	
		Dollars	Percent	Dollars	Percent	Dollars	Percent
Minimum Cost	Mean	4,791,588 (.0035)	9.0% (.0001)	12,315,863 (.0052)	10.2% (.0001)	570,652 (.0005)	8.4% (.0001)
	Median	564,631 (.0001)	6.9% (.0001)	4,219,111 (.0001)	6.5% (.0001)	151,753 (.0001)	7.0% (.0001)
Maximum Cost	Mean	-5,788,561 (.0053)	-9.1% (.0001)	-15,259,686 (.0062)	-12.5% (.0001)	-475,490 (.0058)	-7.1% (.0001)
	Median	-397,922 (.0001)	-5.5% (.0001)	-6,016,445 (.0001)	-10.8% (.0001)	-134,822 (.0001)	-3.8% (.0001)
Closing Price Cost	Mean	17,838 (.3289)	0.3% (.0495)	32,107 (.5227)	0.05% (.4039)	9,834 (.1713)	0.4% (.0633)
	Median	5,656 (.0032)	1.3% (.0005)	5,187 (.5155)	0.04% (.3264)	6,125 (.0015)	0.4% (.0015)
	% > 0	67.2%		56.5%		73.2%	
ASAP Cost	Mean	49,045 (.9750)	1.1% (.4361)	-432,265 (.9223)	0.8% (.7897)	319,048 (.0284)	1.3% (.3942)
	Median	7,360 (.4500)	0.6% (.3704)	-505,677 (.4065)	-1.5% (.6371)	13,565 (.0130)	1.3% (.0570)
	% > 0	54.7%		39.1%		63.4%	
Uniform Cost	Mean	-508,602 (.1374)	-0.4% (.4087)	-1,414,910 (.1383)	-1.5% (.0902)	-185 (.9965)	0.2% (.6470)
	Median	-7,146 (.1909)	-0.2% (.5602)	-580,716 (.0385)	-1.2% (.0188)	2,300 (.1691)	0.3% (.2471)
	% > 0	45.3%		17.4%		61.0%	
Proportional Cost	Mean	-529,147 (.1979)	-0.3% (.5614)	-1,474,062 (.2007)	-1.3% (.1658)	927 (.9818)	0.3% (.5635)
	Median	344 (.4851)	0.0% (.6797)	-490,740 (.0948)	-0.5% (.0573)	8,325 (.0627)	0.6% (.0296)
	% > 0	50.0%		21.7%		65.9%	

Table 8: Bid-Ask Spread Liquidity Effects

Statistics on daily average spread data, aggregated across all firm trade days are reported broken out by exchange listing and corporate percent of daily volume. Panel A contains summary information for the repurchase days with a valid Lag Day comparison. Panels B contains summary information for repurchase days with a lead day against which to benchmark. Each panel presents summary statistics on the daily average dollar and percentage spread and provides the p-values for Wilcoxon Signed Ranks and Student-t statistics against the null hypothesis that the corporate spread is equal to the relevant benchmark.

		Firm Days	Mean		Student		Median		Signed	
			Benchmark Spread	Corp Spread	Mean Change	t-stat p-value	Benchmark Spread	Corp Spread	Median Change	Ranks p-value
Panel A: Lag Day Comparison										
Full Sample	\$	642	.3432	.3347	-.0085	(.0082)	.2500	.2500	.0000	(.0189)
	%		2.01%	1.96%	-0.05%	(.0403)	1.29%	1.25%	-0.04%	(.0883)
NASDAQ	\$	349	.4560	.4448	-.0112	(.0398)	.3750	.3750	.0000	(.0912)
	%		3.16%	3.09%	-0.08%	(.0880)	2.82%	2.75%	-0.07%	(.3678)
NYSE	\$	293	.2210	.2037	-.0173	(.0586)	.1935	.1834	-.0101	(.0591)
	%		0.64%	0.62%	-0.02%	(.0486)	0.59%	0.56%	-0.03%	(.0638)
Panel B: Lead Day Comparison										
Full Sample	\$	642	.3439	.3338	-.0101	(.0042)	.2500	.2422	-.0078	(.0024)
	%		2.00%	1.94%	-0.06%	(.0345)	1.26%	1.22%	-0.04%	(.0332)
NASDAQ	\$	350	.4573	.4447	-.0127	(.0381)	.3609	.3290	-.0319	(.0332)
	%		3.13%	3.05%	-0.09%	(.0905)	2.76%	2.65%	-0.10%	(.1654)
NYSE	\$	292	.2079	.2008	-.0071	(.0077)	.1919	.1829	-.0090	(.0345)
	%		0.64%	0.61%	-0.03%	(.0039)	0.57%	0.56%	-0.02%	(.0397)

*** For 50 firms with identifiable and uncontaminated pre-repurchase horizon trading data available.

Table 9: Price Impact Liquidity Effects

Each day is partitioned into 13 one-half hour trading windows. The dependent variable in the estimation is the return in each time window measured as the change in the mid-point of the bid-ask spread from the beginning of the period to the end of the period. We estimate two regressions. The first isolates repurchase impact under periods of sell pressure. The second regression isolates repurchase impact under periods of buy pressure. The independent variables are the net turnover (NTO) measured as the buy volume during the period less the sell volume during the period, scaled by shares outstanding, and three indicator variables. We code (Neg = 1) for periods of negative NTO and zero otherwise. We code (Pos = 1) for periods of positive NTO and zero otherwise. We also include an indicator set to one (R = 1) for days of successful repurchase. (P-values are reported in parentheses).

	Nyse Estimations		NASDAQ Estimations	
Intercept (α_1)	0.0665 (.0001)	-0.1046 (.0001)	0.0127 (.0002)	-0.0123 (.0001)
NTO (β_1)	2.2023 (.0001)	1.1674 (.0001)	0.3472 (.0001)	0.0596 (.0343)
Negative (α_2)	-0.1853 (.0001)		-0.0249 (.0001)	
NTO * Neg (β_2)	-1.1042 (.0001)		-0.1688 (.0062)	
R * Neg (α_3)	-0.0009 (.9069)		-0.0001 (.9871)	
R * Neg * NTO (β_3)	-0.0446 (.8025)		-0.1734 (.0051)	
Positive (α_2)		0.1784 (.0001)		0.0186 (.0003)
NTO * Pos (β_2)		1.9536 (.0001)		0.6344 (.0001)
R * Pos (α_3)		0.0000 (.9968)		0.0188 (.0427)
R * Pos * NTO (β_3)		-1.6019 (.0001)		-0.6185 (.0001)
F Value	452.4	447.2	37.6	52.2
Prob > F	(.0001)	(.0001)	(.0001)	(.0001)
Adj. R-square	0.0581	0.0575	0.0084	0.0118

Price Support, Random Trading or Timing Skill?

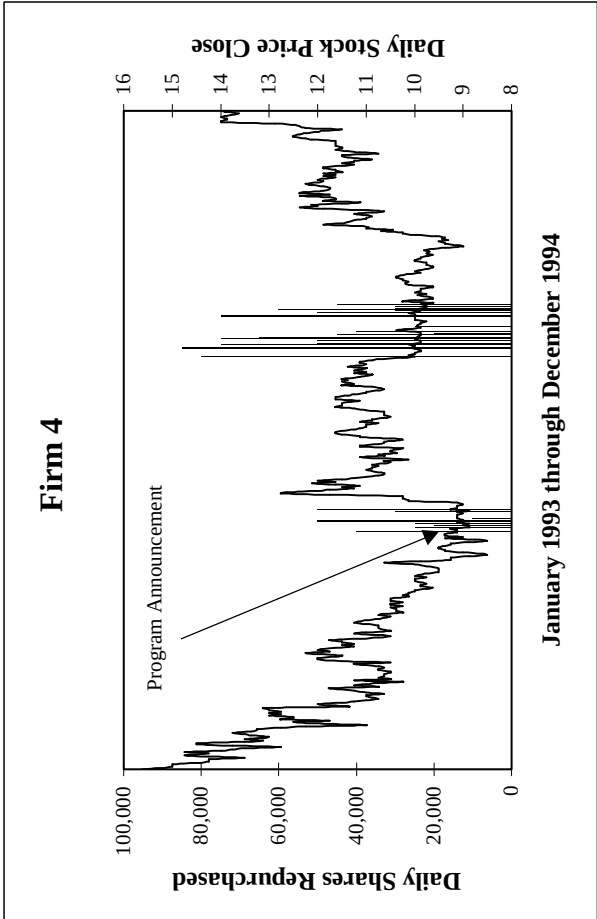
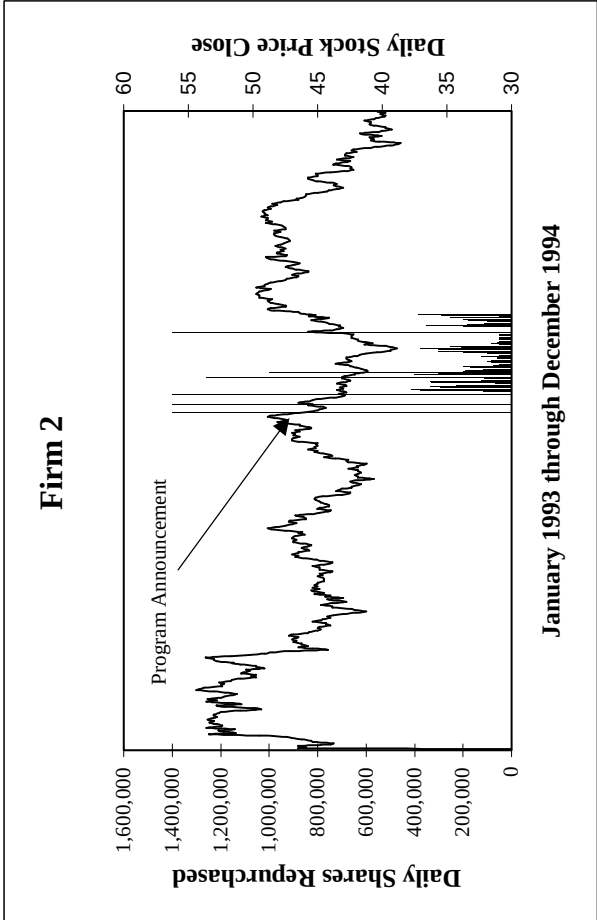
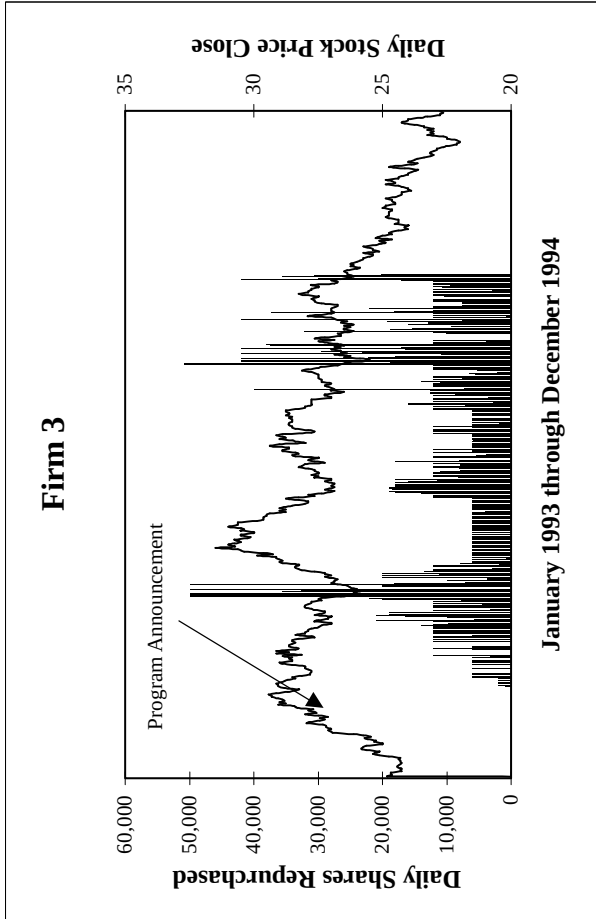
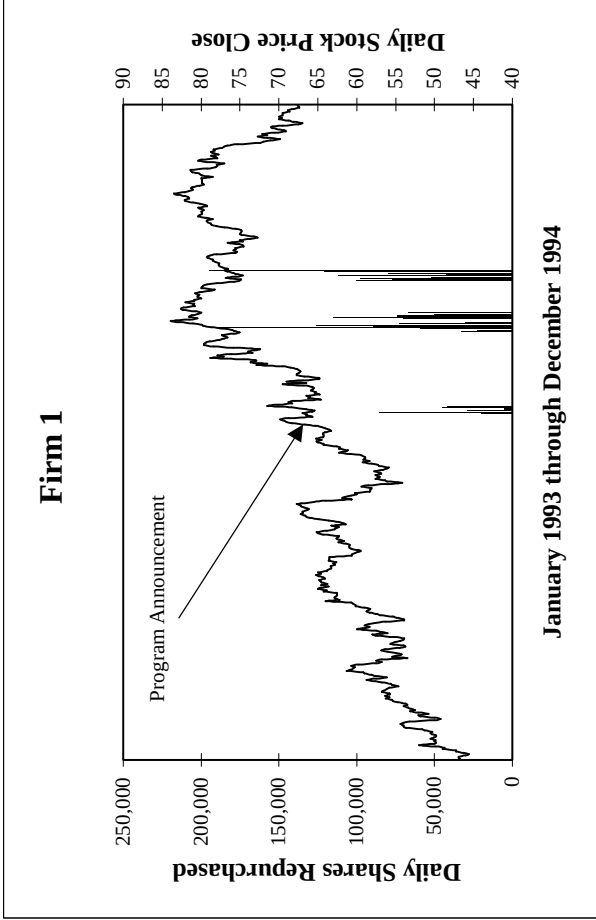
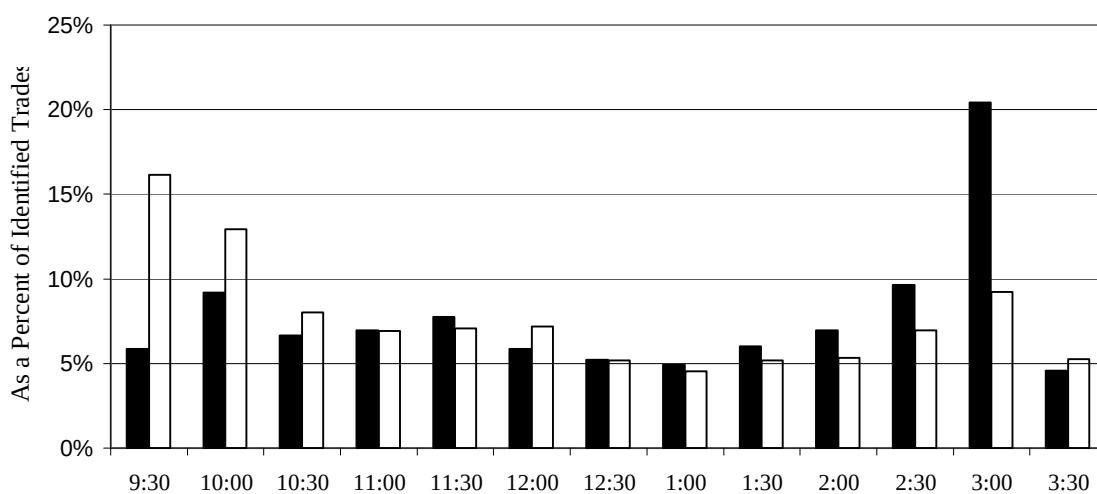


Figure 1
Daily closing stock price and shares repurchased are presented for four representative firms over the period January 1993 through December 199

Corporate Trading by the Half Hour



Aggression of Repurchases

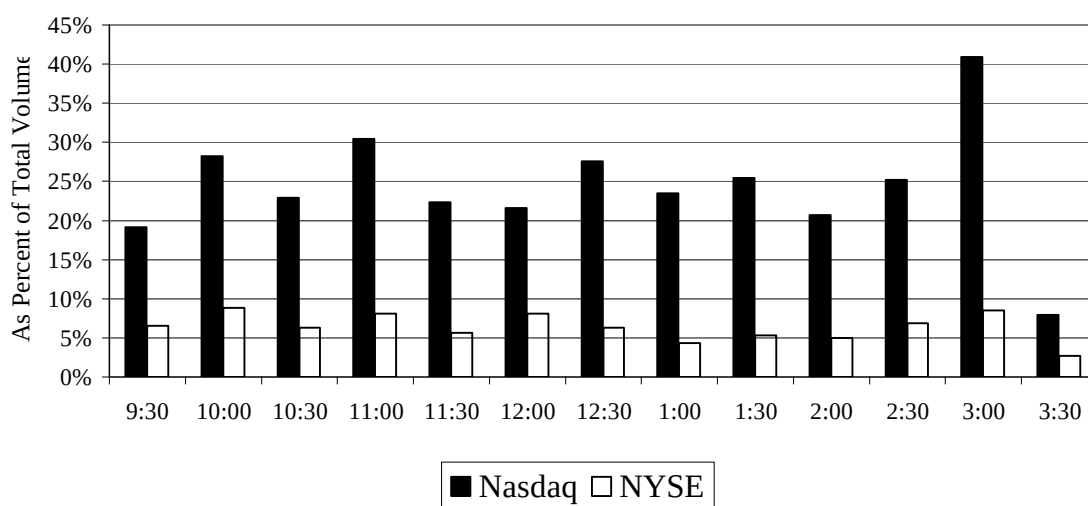
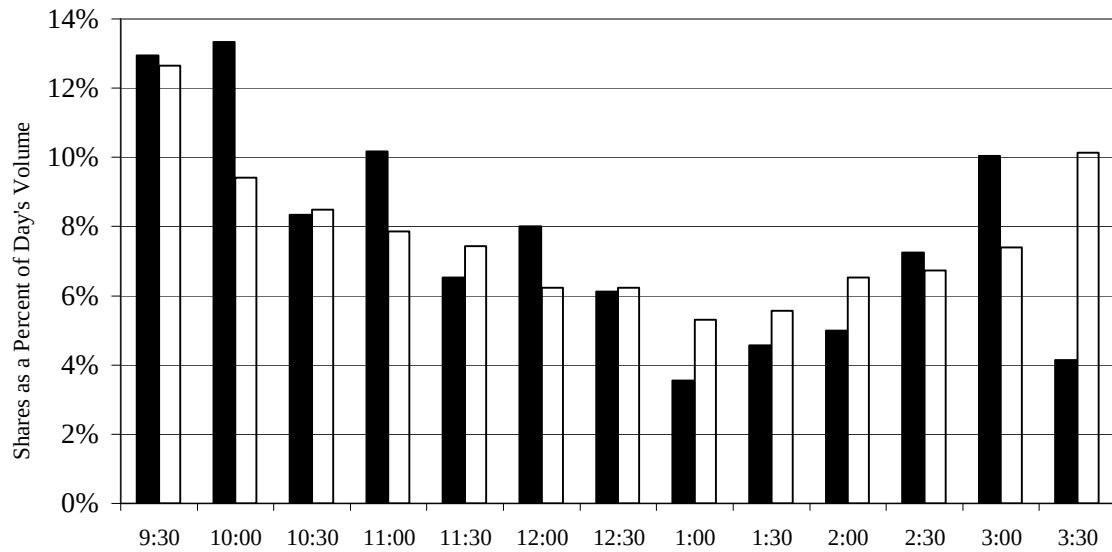


Figure 2

Identified Repurchase shares are presented for Nasdaq and NYSE firms partitioned by trading time. Trading is broken down into half-hour intervals. In the bottom graph, identified repurchases are displayed as a percent of total executed trading volume. Note that the total volume is inclusive of unidentified repurchases.

NYSE Trading by Half Hour



Nasdaq Trading by Half Hour

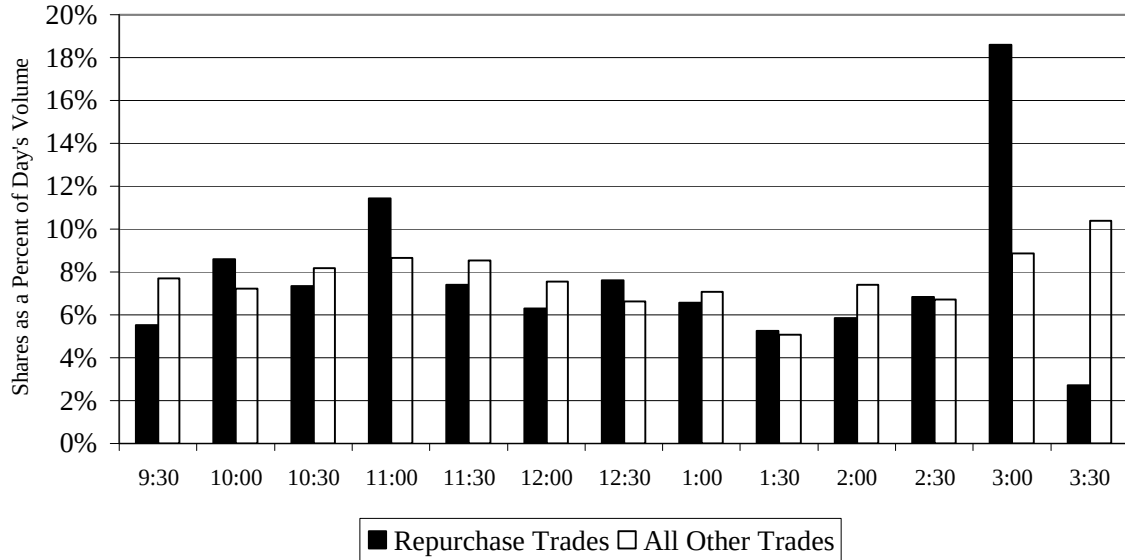


Figure 3

Identified Repurchase shares and non-corporate volume (which includes unidentified repurchase shares) are presented for Nasdaq and NYSE firms separately. Trading is broken down into half-hour intervals.

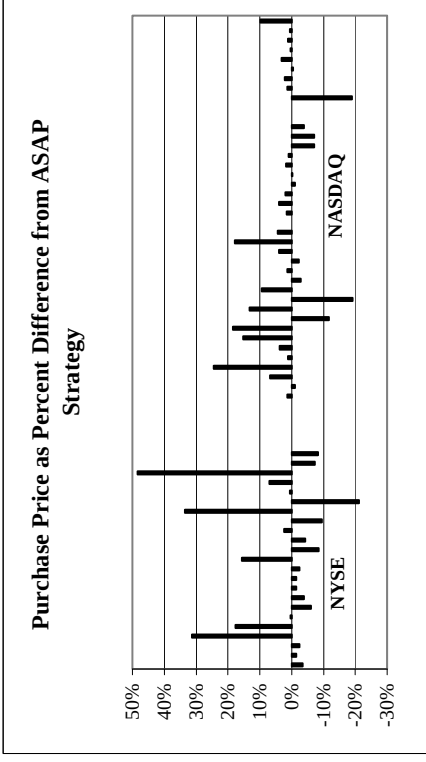
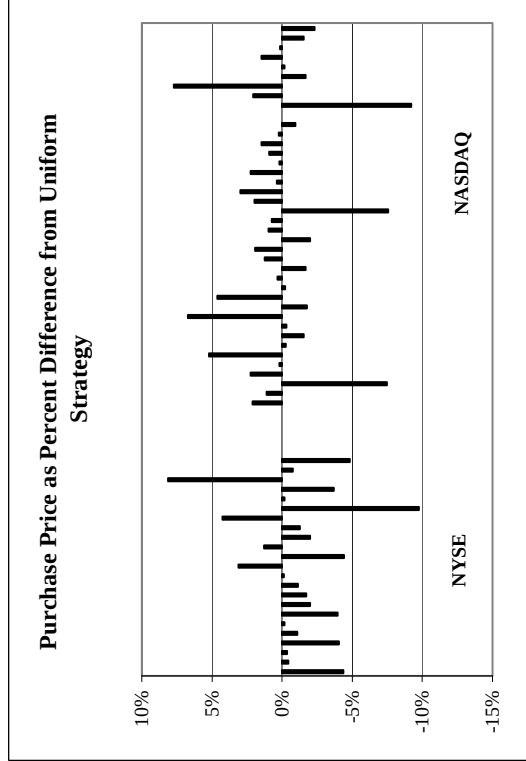
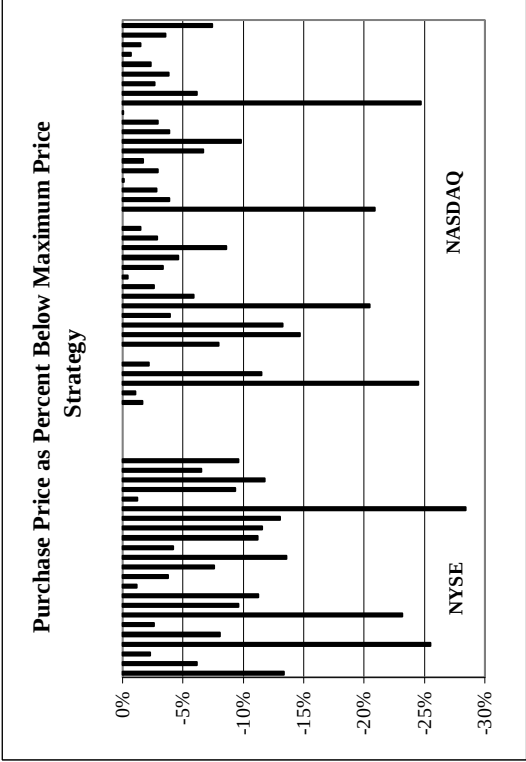
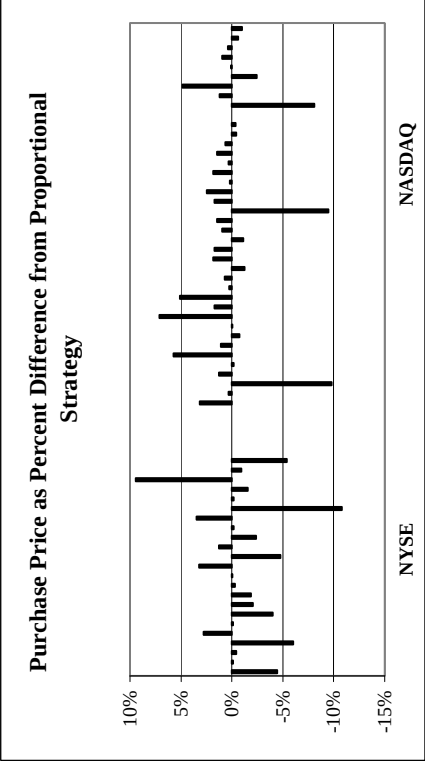
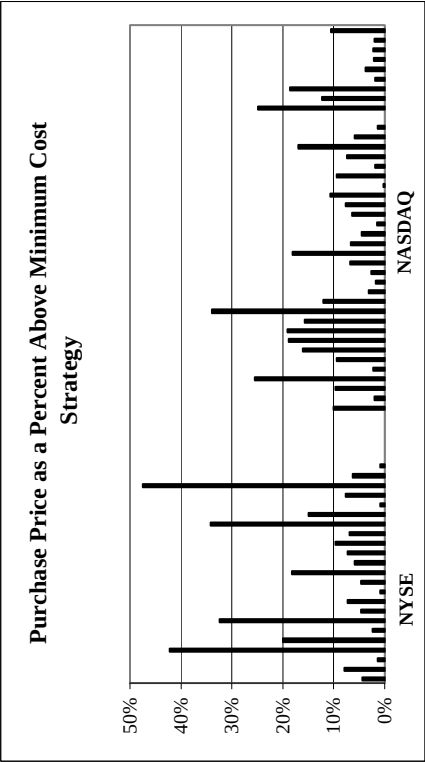


Figure 4

The actual repurchase cost for each of the 64 programs in our study are presented as a percent of five different cost-based benchmarks; uniform, proportional, as soon as possible, minimum and maximum cost.

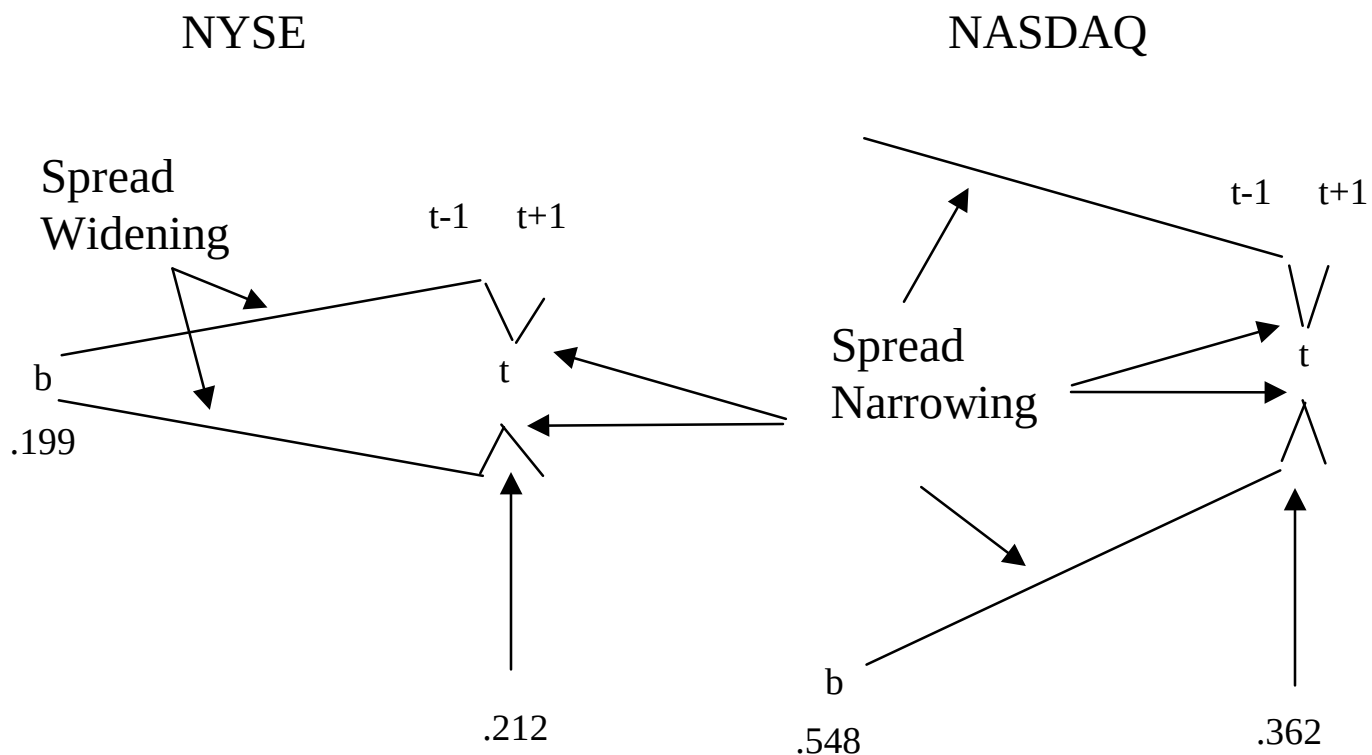


Figure 5: The change in bid-ask spread is represented for paired differences starting at the pre-announcement benchmark horizon (b), through the lag day benchmark ($t-1$), the day before the repurchase day (t), and the lead benchmark ($t+1$) following the repurchase day. We find evidence of a spread widening on the NYSE relative to the pre-announcement period, and spread narrowing relative to lead and lag day benchmarks. On the Nasdaq, we find only evidence of spread narrowing.